Q51 A patient presents with severe dehydration. If the arterial blood pH is 7.4 and the concentration of bicarbonate is twice the concentration of dissolved carbon dioxide, what is the log of the ratio of conjugate base to acid according to the Henderson-Hasselbalch equation?

a. 1.0

b. 0.3

c. 0.7

d. 0.0

e. 2.0

Q52 The high boiling point, high viscosity, and high surface tension of water are critical for maintaining the thermal stability of the human body. These unique physical properties are primarily a result of:

a. Ionic interactions

b. Hydrogen bonding

c. Van der Waals forces

d. Covalent bonding

e. Hydrophobic effects

Q53 In the fluid mosaic model of biological membranes, which molecule is responsible for intercalating between phospholipids to regulate membrane permeability and maintain fluidity?

a. Integral protein

b. Peripheral protein

c. Cholesterol

d. Glycolipid

e. Sphingomyelin

Q54 Which type of transmembrane transport moves a solute against its electrochemical gradient by coupling the movement directly to the hydrolysis of adenosine triphosphate?

a. Simple diffusion

b. Facilitated diffusion

c. Primary active transport

d. Secondary active transport

e. Osmosis

Q55 A 5-year-old child presents with a history of multiple bone fractures following minor trauma and a distinct blue tint in the sclera. This clinical presentation is most characteristic of a genetic defect in:

a. Elastin

b. Fibronectin

c. Laminin

d. Type I Collagen

e. Chondroitin sulfate

Q56 Which glycosaminoglycan is found abundantly in the cornea and cartilage, and is unique because its carbohydrate chain contains galactose instead of uronic acid?

a. Hyaluronic acid

b. Chondroitin sulfate

c. Heparin

d. Keratan sulfate

e. Dermatan sulfate

Q57 Proteins that act as biological catalysts to lower the activation energy of a chemical reaction without being consumed in the process are classified as:

a. Hormones

b. Receptors

c. Structural proteins

d. Antibodies

e. Enzymes

Q58 Albumin and Hemoglobin are examples of globular proteins that maintain oncotic pressure and deliver respiratory gases, respectively. They are primarily classified by function as:

a. Transport proteins

b. Contractile proteins

c. Storage proteins

d. Regulatory proteins

e. Catalytic proteins

Q59 A newborn displays signs of maple syrup urine disease. The biochemical basis of this condition involves the inability to decarboxylate branched-chain amino acids. Which of the following is an affected amino acid?

a. Leucine

b. Glycine

c. Arginine

d. Phenylalanine

e. Histidine

Q60 Tyrosine is a non-essential amino acid that becomes "conditionally essential" in patients suffering from which of the following metabolic disorders?

a. Albinism

b. Phenylketonuria

c. Alkaptonuria

d. Cystinuria

e. Homocystinuria

Q61 Which of the following amino acids contains an imidazole ring in its side chain, allowing it to function as a significant physiological buffer at a pH of 7.4?

a. Lysine

b. Arginine

c. Histidine

d. Tryptophan

e. Proline

Q62 Which amino acid is considered strictly ketogenic, as its metabolic catabolism exclusively produces Acetyl-CoA or Acetoacetate?

a. Alanine

b. Leucine

c. Serine

d. Valine

e. Glycine

Q63 Which amino acid possesses a secondary amino group (imino group) that often leads to the termination of an alpha-helix, causing a structural kink?

a. Proline

b. Glycine

c. Alanine

d. Leucine

e. Isoleucine

Q64 The folding of a single polypeptide chain into its final three-dimensional shape, stabilized by disulfide bonds, salt bridges, and hydrophobic interactions, is known as the:

a. Primary structure

b. Secondary structure

c. Tertiary structure

d. Quaternary structure

e. Supersecondary structure

Q65 During the process of protein denaturation, which of the following chemical bonds remains intact while the higher-order structures are lost?

a. Hydrogen bonds

b. Hydrophobic interactions

c. Peptide bonds

d. Ionic bonds

e. Van der Waals forces

Q66 A patient is diagnosed with Methemoglobinemia. The hemoglobin in their blood is unable to release oxygen to peripheral tissues because the iron atom in the heme group is in which state?

a. Ferrous (Fe^{2+})

b. Ferric (Fe^{3+})

c. Elemental (Fe^{0})

d. Cuprous (Cu^{+})

e. Cupric (Cu^{2+})

Q67 The "Tense" (T) state of hemoglobin is characterized by low oxygen affinity and is primarily stabilized by the ionic binding of:

a. Carbon monoxide

b. 2,3-Bisphosphoglycerate

c. Nitric oxide

d. Myoglobin

e. Glucose

Q68 Which plasma protein is a positive acute-phase reactant that increases significantly in concentration during tissue injury, infection, or acute inflammation?

a. Albumin

b. C-reactive protein

c. Transferrin

d. Antithrombin

e. Prealbumin

Q69 Which immunoglobulin is found in high concentrations in breast milk and saliva, providing the primary line of immunological defense at mucosal surfaces?

a. IgG

b. IgA

c. IgM

d. IgE

e. IgD

Q70 A patient with a parasitic infection and severe extrinsic asthma is likely to show a significant elevation in which immunoglobulin class?

a. IgG

b. IgA

c. IgM

d. IgE

e. IgD

Q71 A distinct "spike" in the gamma-globulin region of a serum protein electrophoresis (SPEP) result is most indicative of which pathology?

a. Liver Cirrhosis

b. Multiple Myeloma

c. Nephrotic Syndrome

d. Iron Deficiency Anemia

e. Vitamin C Deficiency

Q72 A patient with Wilson's disease shows a characteristic decrease in the serum levels of which copper-carrying plasma protein?

a. Albumin

b. Haptoglobin

c. Ceruloplasmin

d. Transferrin

e. Hemopexin

Q73 Which of the following lipids is the major structural component of the lipid bilayer and possesses an amphipathic nature?

a. Triglyceride

b. Cholesterol ester

c. Phospholipid

d. Wax

e. Squalene

Q74 Essential fatty acids, such as Linoleic acid, are required in the human diet because the body lacks the enzymes to introduce double bonds beyond which carbon atom?

a. Carbon 3

b. Carbon 6

c. Carbon 9

d. Carbon 12

e. Carbon 15

Q75 Which lipoprotein is responsible for the transport of cholesterol from the liver to the peripheral tissues and is clinically associated with atherosclerosis?

a. Chylomicron

b. VLDL

c. LDL

d. HDL

e. IDL

Q76 What is the specific oxidation state of the iron atom within a functional, oxygen-binding heme group of hemoglobin?

a. Fe0

b. Fe2+

c. Fe3+

d. Fe4+

e. Fe1+

Q77 Which fat-soluble vitamin is required as a cofactor for the gamma-carboxylation of specific glutamate residues in several blood clotting factors?

a. Vitamin A

b. Vitamin D

c. Vitamin E

d. Vitamin K

e. Vitamin C

Q78 Which of the following is the predominant intracellular cation responsible for maintaining the resting membrane potential?

a. Sodium

b. Potassium

c. Calcium

d. Magnesium

e. Iron

Q79 Which cellular organelle is the primary site for the synthesis of steroid hormones from cholesterol precursors?

a. Rough Endoplasmic Reticulum

b. Smooth Endoplasmic Reticulum

c. Golgi Apparatus

d. Lysosome

e. Nucleus

Q80 During DNA replication, which enzyme is responsible for the ATP-dependent unwinding of the double helix at the replication fork?

a. DNA Ligase

b. DNA Helicase

c. DNA Polymerase

d. DNA Primase

e. DNA Gyrase

Q81 The deficiency of Niacin (Vitamin B3) results in the clinical condition known as:

a. Scurvy

b. Rickets

c. Pellagra

d. Beriberi

e. Pernicious Anemia

Q82 Which type of non-covalent bond primarily stabilizes the repeating helical structure of the protein alpha-helix?

a. Peptide bond

b. Hydrogen bond

c. Ionic bond

d. Disulfide bond

e. Hydrophobic interaction

Q83 In humans, the metabolic end product of purine nucleotide degradation that is excreted in the urine is:

a. Urea

b. Ammonia

c. Uric acid

d. Creatinine

e. Alanine

Q84 Which specific phospholipid acts as a major component of lung surfactant to reduce surface tension within the alveoli?

a. Cholesterol

b. Dipalmitoylphosphatidylcholine

c. Phosphatidylethanolamine

d. Sphingomyelin

e. Cardiolipin

Q85 Under normal physiological conditions, the pH of arterial blood is strictly maintained within the range of:

a. 6.8 - 7.0

b. 7.0 - 7.2

c. 7.35 - 7.45

d. 7.8 - 8.0

e. 7.5 - 7.6

Q86 Which amino acid is the simplest in structure and is the only one that does not possess an asymmetric carbon atom?

a. Alanine

b. Glycine

c. Serine

d. Valine

e. Proline

Q87 The biochemical reactions of the Urea Cycle take place primarily in which organ?

a. Liver

b. Kidney

c. Muscle

d. Brain

e. Lung

Q88 On the basis of its structural organization, Myoglobin is best described as a:

a. Tetrameric protein

b. Monomeric protein

c. Dimeric protein

d. Fibrous protein

e. Multimeric protein

Q89 The enzyme HMG-CoA reductase is the committed, rate-limiting step in the biosynthetic pathway of:

a. Ketone bodies

b. Cholesterol

c. Fatty acids

d. Glycogen

e. Triglycerides

Q90 Which of the following nitrogenous bases is uniquely found in RNA molecules and is absent in DNA?

a. Adenine

b. Guanine

c. Cytosine

d. Thymine

e. Uracil

Q91 The proenzyme trypsinogen, secreted by the pancreas, is converted into its active form, trypsin, by the action of:

a. Pepsin

b. Enterokinase

c. Hydrochloric acid

d. Chymotrypsin

e. Lipase

Q92 The structure of Vitamin B12 (Cobalamin) is unique among vitamins because it contains which central metal ion?

a. Iron

b. Magnesium

c. Cobalt

d. Zinc

e. Copper

Q93 Which of the following is an example of a five-carbon (pentose) sugar involved in the structure of nucleotides?

a. Glucose

b. Fructose

c. Ribose

d. Galactose

e. Mannose

Q94 The specific linear sequence of amino acids in a polypeptide chain constitutes its:

a. Primary structure

b. Secondary structure

c. Tertiary structure

d. Quaternary structure

e. Supersecondary structure

Q95 Which vitamin is a precursor for the retinal pigment rhodopsin and is essential for maintaining normal vision?

a. Vitamin A

b. Vitamin B

c. Vitamin C

d. Vitamin D

e. Vitamin E

Q96 Lesch-Nyhan syndrome is a hereditary disorder of purine metabolism caused by the deficiency of:

a. Glucose-6-phosphate dehydrogenase

b. Hypoxanthine-guanine phosphoribosyltransferase

c. Adenosine deaminase

d. Phenylalanine hydroxylase

e. Hexosaminidase A

Q97 Which of the following is a disaccharide composed of glucose and fructose units?

a. Glucose

b. Sucrose

c. Fructose

d. Galactose

e. Ribose

Q98 The primary structural building blocks of all proteins are:

a. Fatty acids

b. Nucleotides

c. Amino acids

d. Monosaccharides

e. Glycerol

Q99 In eukaryotic cells, the process of genomic DNA replication occurs primarily within the:

a. Cytosol

b. Nucleus

c. Mitochondria

d. Ribosomes

e. Golgi Apparatus

Q100 Unconjugated bilirubin is transported through the circulatory system to the liver bound to which protein?

a. Globulin

b. Albumin

c. Fibrinogen

d. Transferrin

e. Hemoglobin